WIRSCHING'S WEAK COVERING CONJECTURE: AN EXTENDED COMPUTATION, A PROVEN BOUND, AND THE STRUCTURE OF WHAT RESISTS IT

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Abstract. Wirsching's 1998 Weak Covering Conjecture asks for a sub-exponential $K(l)$ such that a large enough set $R_{j-1,j} \subset \mathbb{Z}_3^*$, built from strictly decreasing exponent tuples, is forced to cover every unit residue modulo 3^l. We extend the exact computation of $j^*(l)$, the smallest such covering budget, from $l = 20$ to $l = 23$, and compare four growth models for $e(l) := j^*(l) - l \log_4 3$ against the extended table: a constant model is disfavored at every statistic checked, but logarithmic, square-root, and slow-linear growth remain statistically indistinguishable from one another. We give the best unconditional bound known to us, $j^*(l) \le \frac{119}{104} l + O(1)$, via a mean-payoff-game relaxation of the covering budget. We then show why the natural Fourier approach to a tighter bound cannot work: a uniform square-root-cancellation argument is capped below even this bound, the bulk of the relevant exponential sum's mass sits in an exponentially numerous population no sparse-exception argument reaches, and the residues that actually fail to be covered sit at extremal, non-generic arithmetic cells no averaging method sees. In their place we prove a genuine structural theorem: the set of uncovered residues at one budget is contained in the union of its double and its quadruple at the next, giving an explicit recursive upper bound on $j^*(l)$ that is empirically exact at every budget checked. We falsify a natural strengthening of this theorem with an explicit witness-level counterexample, prove three further exact laws governing the last residues to be covered, and name precisely the two lemmas whose proof would either sharpen this bound further or show it cannot be. The Weak Covering Conjecture itself remains open.

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1. Introduction

Wirsching's 1998 monograph [1] studies the density of predecessor sets under the $3n + 1$ map through an averaging construction on the 3-adic integers. A limiting Markov chain built from that construction has an invariant density, and Wirsching's own route to positive predecessor density passes through a covering question about a family of finite subsets of $\mathbb{Z}_3^*$, the 3-adic units. For integers $j, k \ge 0$, write

$$R_{j,k} := \left\{ \sum_{i=0}^{j} 2^{a_i} 3^i \;\middle|\; j + k \ge a_0 > a_1 > \cdots > a_j \ge 0 \right\} \subset \mathbb{Z}_3^*. \tag{1}$$

Each element of $R_{j,k}$ is built from a strictly decreasing sequence of $j + 1$ "digit positions" chosen from $\{0, \ldots, j + k\}$: the i-th chosen position, counting from the top, contributes 2^{a_i} scaled by 3^i. Wirsching's Weak Covering Conjecture asks for a sub-exponential $K(l)$ such that $|R_{j-1,j} \bmod 3^l| \ge K(l) \cdot 3^l$ already forces $R_{j-1,j}$ to cover every unit residue modulo 3^l. Equivalently, writing $j^*(l)$ for the smallest j such that $R_{j-1,j}$ covers $(\mathbb{Z}/3^l\mathbb{Z})^*$ exactly, the conjecture concerns the growth rate of

$$e(l) := j^*(l) - l \log_4 3, \tag{2}$$

the excess of the true covering budget over the leading-order rate a cardinality argument alone predicts ($|R_{j-1,j}| = \binom{2j}{j} \sim 4^j$, so matching $2 \cdot 3^{l-1}$ residues needs $j \sim l \log_4 3$ at leading order).

An earlier, unpublished manuscript of the author computed $j^*(l)$ exactly for $l = 1, \ldots, 20$ by direct enumeration and reported the resulting table without a growth-rate analysis. This paper does three things. First, it extends that table to $l = 23$ and gives a proper statistical comparison of candidate growth models for $e(l)$ against the full range (Sections 2 and 3). Second, it gives the best unconditional upper bound on $j^*(l)$ we have found, via a reformulation of the covering budget as a mean-payoff game (Section 4). Third, and at greatest length, it reports a sustained attempt to close the gap to the conjectured rate directly, first by the natural Fourier approach, and, once that approach is shown to fail for a precise and checkable reason, by a direct study of the residues that actually resist coverage (Sections 5 and 6). That second attempt does not settle the conjecture. It produces a genuine theorem, a genuine falsification of a natural strengthening of that theorem, and two precisely named open lemmas, each of which we believe is more tractable than the conjecture itself and worth a dedicated attack in its own right.

2. The computation

Table 1 gives $j^*(l)$ and $e(l)$ for $l = 1, \ldots, 23$. The values for $l \leq 20$ reproduce the earlier manuscript's table exactly; they were independently recomputed here by a from-scratch Rust reimplementation of the covering search (a bitset rotation dynamic program over residues modulo 3^l), validated against the original Python implementation at every $l \leq 20$ before being trusted for the extension. The values for $l = 21, 22, 23$ are new.

l	$j^*(l)$	$e(l)$	l	$j^*(l)$	$e(l)$	l	$j^*(l)$	$e(l)$
1	1	0.208	9	13	5.868	17	21	7.528
2	4	2.415	10	15	7.075	18	22	7.735
3	6	3.623	11	16	7.283	19	23	7.943
4	7	3.830	12	17	7.490	20	24	8.150
5	9	5.038	13	18	7.698	21	25	8.358
6	10	5.245	14	19	7.905	22	26	8.565
7	11	5.453	15	20	8.113	23	27	8.773
8	12	5.660	16	20	7.320			

TABLE 1. $j^*(l)$ and $e(l) = j^*(l) - l \log_4 3$, $l = 1, \ldots, 23$.

The single plateau in the table, $j^*(15) = j^*(16) = 20$, is not an error: $l = 15 \to 16$ is the only increment among the 22 steps from $l = 1$ to $l = 23$ at which the covering budget does not increase, and it is exactly the event Section 3's plateau-frequency test measures.

$l = 21$ fits in physical RAM once a memory-wasteful parallelization step in the original implementation was corrected. $l = 22$ required 500 GiB of swap, added to the machine specifically for this computation, and took approximately 10,749 s (~ 3 hours) with swap I/O as the dominant cost, against 748 s for $l = 21$ without swap. $l = 23$ required approximately 263 GiB of state and 375,615 s ($\sim$104 hours); the successful run had no interruption, so a checkpoint/resume mechanism added after an earlier, unrelated policy-triggered kill of an interrupted attempt at the same level was never exercised on it. Three attempts at $l = 24$, made over the course of this project on the same machine, each failed before completion (two to memory-policy kills, one to an apparent out-of-memory condition with no surviving kernel log confirming it directly); $l = 24$ is not attempted further, and the table stops at $l = 23$.

3. The growth of $e(l)$

We compare four candidate functional forms for $e(l)$'s growth against the tail $l = 10, \ldots, 23$ ($n = 14$; the same range choice as the earlier manuscript, now with three more points): a constant (stabilization), a logarithmic term, a square-root term, and a slow-linear term. Table 2 gives the fit statistics.

model	k	ΔAIC	ΔBIC	LOOCV RMSE
constant	1	16.091	15.452	0.5200
logarithmic	2	1.512	1.512	0.2991
square-root	2	0.722	0.722	0.2909
slow-linear	2	0.000	0.000	0.2837

Table 2. Model comparison on $e(l)$, $l = 10, \dots, 23$. ΔAIC/BIC relative to the best-fitting model (slow-linear).

A constant model is disfavored sharply and consistently: $\Delta\text{AIC} = 16.09$, $\Delta\text{BIC} = 15.45$, both growing at every l added since $l = 21$. The three growth models remain statistically indistinguishable from one another by the conventional $\Delta\text{AIC} > 2$ rule of thumb, with slow-linear taking a narrow lead on both AIC and leave-one-out cross-validation; the three candidate regressors ($\log l$, $\sqrt{l}$, l) are themselves highly correlated over this range (pairwise correlation ≥ 0.993), so this ranking says more about how thin the margin currently is than about which functional form is correct.

A direct combinatorial test tells the same story from a different angle. Among the 22 increments $j^*(l+1) - j^*(l)$ for $l = 1, \dots, 22$, exactly one is zero (the $l = 15 \to 16$ plateau noted above). A one-sided binomial test against the plateau probability a constant-$e(l)$ source would imply gives $p = 0.0426$, crossing the conventional 0.05 threshold for the first time with the $l = 23$ point included; restricted to the same tail the model comparison above uses, the same test gives $p \approx 0.22$. We report both figures rather than only the more dramatic pooled one: $e(l)$ is an exact, deterministic sequence, not a sample from a noisy process, so neither AIC/BIC nor this test carries the sampling interpretation of a classical inference. Both are best read descriptively, as "how well does a constant-$e(l)$ source fit the observed increment pattern," not as a formal hypothesis test.

A constant $e(l)$ is not merely disfavored statistically; it is impossible.

Proposition 1 (Unboundedness of $e(l)$)**.** $e(l) \geq \frac{1}{4}\log_2 l - O(1)$.

Proof. $R_{j-1,j}$ is, by the bijection between strictly decreasing exponent tuples in $\{0, \dots, 2j-1\}$ and j-subsets of a $2j$-element set, of exact cardinality $\binom{2j}{j}$. Covering $(\mathbb{Z}/3^l\mathbb{Z})^*$, of size $2 \cdot 3^{l-1}$, requires the domain to be at least as large as the codomain: $\binom{2j}{j} \geq 2 \cdot 3^{l-1}$ is necessary for $j \geq j^*(l)$. The standard asymptotic $\binom{2j}{j} \sim 4^j/\sqrt{\pi j}$ inverts this inequality to $j \geq l \log_4 3 + \frac{1}{4}\log_2 l - O(1)$. $\square$

This argument uses no arithmetic structure of $R_{j,k}$ beyond its cardinality, and a working number theorist would write it down without pausing on it; we verified it three ways before including it (by hand, against the exact table via direct integer search with zero violations at every $l \leq 23$, and by an independent, cold rederivation from a second reviewer) precisely because it is otherwise the kind of one-line claim that is easy to get an exponent wrong on. We record it as a rigorous floor under the statistical picture above, not as progress toward the true rate.

4. A mean-payoff-game upper bound

The covering budget admits an exact reformulation as a game of perfect information. A state is a unit $z \in \mathbb{Z}_3^*$; a move of cost $d \geq 0$ is legal at z exactly when $2^d z \equiv 1 \pmod 3$, and it replaces z with $T_d(z) := (2^{d+1}z - 2)/3$. $j^*(l)$ is the value, over l steps with the modulus shrinking by one power of 3 per step, of the max-over-z, min-over-policy total cost.

Restricting a policy to see only $z \bmod 3^k$ turns this infinite-information problem into a finite, two-player *mean-payoff game*: the minimizing player chooses a legal d knowing only $z \bmod 3^k$, an adversary then chooses the hidden next 3-adic digit $\epsilon \in \{0, 1, 2\}$, and the state updates to

$T_d(z+3^k\epsilon) \bmod 3^k$. By the classical theorem of Ehrenfeucht and Mycielski [2], every mean-payoff game on a finite graph has a value at each state, securable by both players with memoryless (positional) strategies; here the value is independent of the starting state and we write it ρ_k.

Theorem 2 (Window relaxation bound). *For every $k \geq 1$ there is an explicit rational constant C_k such that $j^*(l) \leq \rho_k \cdot l + C_k$ for all l, and ρ_k is non-increasing in k.*

ρ_k and C_k are computed exactly, as rationals, by a nested Howard strategy-improvement solver, self-certified at every k via a matching adversary lower bound extracted from the same run. Table 3 gives the values computed for $k = 3, \ldots, 13$.

k	3	4	5	6	7	8	9	10	11	12	13
ρ_k	2	$\frac{5}{3}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{7}{5}$	$\frac{25}{19}$	$\frac{5}{4}$	$\frac{11}{9}$	$\frac{6}{5}$	$\frac{7}{6}$	$\frac{119}{104}$

TABLE 3. The window-relaxation value ρ_k, $k = 3, \ldots, 13$.

At $k = 13$, $\rho_{13} = 119/104 = 1.144231\ldots$, giving:

Corollary 3. $j^*(l) \leq \frac{119}{104} l + O(1)$.

This is, to our knowledge, the best unconditional bound on $j^*(l)$'s growth rate currently established. The general technique, representing a covering-type combinatorial problem as a mean-payoff game and computing a decreasing sequence of exact rational relaxation bounds, is not new to this application: beyond Ehrenfeucht and Mycielski's foundational theorem [2] itself, a close and recent methodological precedent proves rationality and computability of the covering radius of a primitive sofic shift by essentially the same construction, applied to a different object [3]. What is new here is the specific application to $j^*(l)$, not the framing, and we cite both as precedent rather than present the technique itself as a contribution of this paper.

Remark 4. The trailing behaviour of the computed ρ_k sequence, $\rho_k \to L$ with $L \approx 0.78$–0.79 under a naive $L + A/k$ extrapolation, is consistent with the conjectured true rate $\log_4 3 = 0.7925\ldots$, but we emphasize that only the direction $j^*(l) \leq \rho_k l + C_k$ is proven here. The reverse inequality, $\inf_k \rho_k \leq \limsup_l j^*(l)/l$, which would be needed to conclude $\rho_k \to \log_4 3$ rather than merely $\rho_k \to$ some value $\geq \log_4 3$, is not established, and we found evidence in the course of this work that the window relaxation may carry an irreducible gap above the true rate. We flag this explicitly because an earlier internal draft of this result asserted the reverse inequality without proof; it does not hold as stated.

5. The exponential sum, and why it resists a direct attack

Corollary 3's bound, $1.1442\ldots\, l$, is well above the conjectured rate $\log_4(3)\, l \approx 0.7925\, l$. The natural route to closing that gap is Fourier-analytic. By orthogonality on $\mathbb{Z}/3^l\mathbb{Z}$, the number of tuples landing in residue class z is

$$N(z) = \frac{1}{3^l} \sum_{t=0}^{3^l-1} S(t)\, e\!\left(-\frac{tz}{3^l}\right), \qquad S(t) := \sum_{j+k \geq a_0 > \cdots > a_j \geq 0} e\!\left(\frac{t}{3^l} \sum_{i=0}^{j} 2^{a_i} 3^i\right), \tag{3}$$

and a uniform bound $|S(t)| = o(|R_{j,k}|)$ for every nonzero t would give $N(z) > 0$ for every z once $|R_{j,k}|$ is large enough, at a rate controlled directly by the exponent in the bound. This approach does not work, for three separate and mutually reinforcing reasons.

5.1. A hard ceiling on uniform cancellation.

Proposition 5. *Any proof of covering via a uniform square-root-type bound on $|S(t)|$, applied via the triangle inequality across all $3^l - 1$ nonzero frequencies, requires $j \geq 2\log_4(3)\, l \approx 1.585\, l$.*

Proof. Write $T := |R_{j-1,j}| = \binom{2j}{j}$. Positivity of $N(z)$ for every z via $\sum_{t\neq 0} |S(t)| < T$ combined with a uniform estimate $|S(t)| \lesssim \sqrt{\beta T}$ over the $3^l - 1$ nonzero frequencies forces $3^l\sqrt{\beta T} \lesssim T$, i.e. $T \gtrsim 3^{2l}$, i.e. $j \gtrsim 2\log_4(3)\, l$. □

Proposition 5's threshold is *worse* than Corollary 3's already-proven bound: any approach that only delivers generic square-root cancellation across every frequency cannot even match what is already known by other means, let alone improve on it. This rules out the naive version of the Fourier approach cleanly, and directs attention to whether a finer, frequency-dependent treatment (a major/minor-arc split, in the language of the circle method) can do better.

5.2. The exceptional mass does not stay small.

Numerically, $|S(t)|$ is far from uniform: the frequencies of largest magnitude sit extremely close to dyadic rationals of small denominator, $t/3^l \approx p/2^r$ for small r, with the quality of approximation improving by almost exactly a factor of 3 per unit increase in l, the signature of an underlying connection to the classical effective irrationality measure of $\log_2 3$ via transcendence theory (Baker's theorem on linear forms in logarithms). This connects the problem directly to a 2011 discussion of Terence Tao's, applying Littlewood–Offord and Riesz-product methods to an object built from the same $2^a 3^b$-type exponential sums, in the closely related setting of a hypothetical non-trivial Collatz cycle [4]. Reindexing Tao's construction identifies it as an exact boundary slice of $R_{j,k}$ itself, with the bottom exponent pinned at 0; his own conclusion, in his own words, is that "once one uses the transcendence theory bound, ... Littlewood-Offord techniques are no longer of much use," and that he knows "of no way to get a nontrivial separation property between powers of 2 and powers of 3 other than via transcendence theory methods."

Whether the sparse major-arc resonances described above can nonetheless be excised and the remainder controlled by an averaged bound is a separate, checkable question. It fails: at the threshold $\eta \sim c/\sqrt{T}$ that actually governs the L^1 norm (rather than at a fixed threshold, where exceptional frequencies do look sparse), the count of exceptional frequencies grows genuinely exponentially in l, roughly threefold per unit increase, with the proportion exceeding a fixed multiple of $\sqrt{T}$ settling to a stable positive constant. At $l = 18$, summing every one of the 8,014 primitive coefficients exceeding even a generous fixed threshold of $0.001 \cdot T$ contributes less than 132 to an L^1 norm of 5226: over 97% of the mass lives in an exponentially numerous population of near-average-magnitude coefficients, not in the visible spikes. No sparse-exceptional-set repair of the naive approach is available.

5.3. Coverage failure is an arithmetic, not a statistical, phenomenon.

The most direct test is to examine the residues that actually fail to be covered, one budget step before covering succeeds. At every level checked, their count is tiny (single or low double digits) against an ambient occupancy mean in the hundreds, sitting in cells whose local intensity, at every resolution checked, would assign them probability below e^{-20} under any locally homogeneous model; a genuinely independent-points model at the same mean would already have covered the residue set completely. A direct experiment confirms this is a phase phenomenon, not a magnitude phenomenon: holding every $|S(t)|$ fixed while randomizing the phases of conjugate-symmetric pairs collapses the observed extremal statistics to what a generic-phase model predicts; the actual, unscrambled phases do not. Coverage failure at the relevant budget is decided by an exact, highly non-generic arithmetic obstruction, invisible to any method that only controls magnitudes, averages, or low moments of $S(t)$.

Proposition 6. $S(3^{l-1})/\binom{2m}{m} \to 1/\sqrt{3}$ *as l/m grows, for the family $R_{m-1,m}$.*

Proof sketch. $\sum_i 2^{a_i}3^i \equiv 2^{a_0} \pmod 3$, so $S(3^{l-1})/\binom{2m}{m}$ depends only on the limiting distribution of the top exponent's parity, which converges to a fixed, computable mixture of the two nontrivial cube roots of unity. □

Uniform decay across every nonzero frequency, the premise a naive statement of the Fourier approach might suggest, is thus not merely difficult; it is false, exactly, at an explicit and easily described family of frequencies.

6. A theorem on the residues that resist coverage

Write $H(l, j)$ for the set of units modulo 3^l *not* covered by $R_{j-1,j}$, so that $j \geq j^*(l)$ exactly when $H(l, j) = \emptyset$.

Theorem 7 (Holdout doubling inclusion)**.** *For $j \geq l$, $H(l, j+1) \subseteq 2H(l, j) \cap 4H(l, j)$.*

Proof. Shifting every exponent in an admissible tuple up by 1 (respectively 2) produces an admissible tuple at budget $j+1$ whose value is exactly twice (respectively four times) the original. So if z is representable at budget $j+1$ via a tuple all of whose exponents are ≥ 1 (respectively ≥ 2), then $z/2$ (respectively $z/4$) is representable at budget j. Once $j \geq l$, prepending a new smallest exponent equal to j contributes $2^j 3^j \equiv 0 \pmod{3^l}$ for $j \geq l$, so this covers the remaining case and every $z \notin 2R_{j-1,j} \cap 4R_{j-1,j}$ failing to be represented at budget $j+1$ must already fail at budget j after halving or quartering. □

Corollary 8 (Chain contraction)**.** *If $x, 2x, 4x, \ldots, 2^{t-1}x \in H(l, j+1)$ (all reduced modulo 3^l), then $x, 2x, \ldots, 2^t x \in H(l, j)$.*

Corollary 9 (Run-length bootstrap)**.** *Writing* $\operatorname{maxrun}(H)$ *for the length of the longest doubling chain contained in H, $j^*(l) \leq j + \operatorname{maxrun}(H(l, j))$ for every $j \geq l$.*

Proof. By Corollary 8, $\operatorname{maxrun}(H(l, j))$ strictly decreases by at least 1 at every budget step until it reaches 0, at which point H is empty. □

Empirical Result 10. Corollary 9 is empirically exact, $j^*(l) = j + \operatorname{maxrun}(H(l, j))$, at every (l, j) with $j \geq l+1$ we were able to compute, $l = 5, \ldots, 20$ at every budget and l up to 22 at the single budget $j = l+1$, including a case with $|H(l, j)| > 3 \times 10^6$.

The converse direction, that a doubling chain of the observed maximal length is not merely sufficient but necessary for survival to that budget, is not proven, and is false as an unrestricted statement: $1547 \in 2H(7, 8) \cap 4H(7, 8)$, yet $1547 \notin H(7, 9)$ (both of 1547's two budget-9 witnesses span the full exponent range, the one case Theorem 7's argument does not directly control). We return to a restricted, still-open form of this converse in Section 7.

A natural strengthening of Theorem 7, that every individual witness admits a bounded-cost local repair, is also false.

Proposition 11 (Failure of uniform local repair)**.** *There is no bounded Δd such that every witness of a covered residue at Durfee depth d lifts to a witness at depth at most $d + \Delta d$ for every residue newly required at the next budget. Explicit witness-level exhaustive checks at two consecutive budget transitions find, respectively,* 100 *of* 1458 *and* 332 *of* 4374 *children whose minimum achievable depth exceeds their parent's minimum depth by more than one, up to three at the transitions checked; a correspondingly exhaustive check of minimum-edit-distance witness repair finds distances up to* 7 *at both transitions, against a distance of* 1 *for the overwhelming majority.*

This falsifies, in its originally proposed strong form, a natural conjecture this project had earlier entertained (that coverage propagates via a uniform, bounded-cost local repair rule budget by budget); its valid surviving content is Theorem 7 and its corollaries above.

7. Further structure of the last holdouts

The residues in $H(l, j^*(l)-1)$, the last to be covered, satisfy several further exact laws.

Theorem 12 (Last-holdout parity)**.** *Every element of* $H(l, j^*(l)-1)$ *is* $\equiv 1 \pmod 3$.

Theorem 13 (Near-extinction bijection)**.** $H(l, j^*(l)-1) = 2\cdot\{x \in H(l, j^*(l)-2) : x \equiv 2 \pmod 3\}$, *exactly.*

Both theorems follow from an exact reparameterization of the covering problem by *width* $W := j+l-1$ rather than by budget j and level l separately: coverage at (l,j), $j \geq l$, depends on (l,j) only through W, via a monotone family of subsets of $\{0,\dots,W\}$; at the minimal width admitting a given residue, every admissible witness is forced to place its largest exponent at the same fixed position, whose parity determines the residue's class mod 3 directly.

Corollary 14 ($\times 4$-only inheritance)**.** *A last holdout at level* $l+1$ *is, if it descends from a last holdout at level* l *at all, related to it by multiplication by* 4*, never by* 2*.*

Proof. By Theorem 7, a last holdout at level $l+1$ pulls back via halving or quartering to $H(l, j^*(l+1)-2)$. By Theorem 12 the level-$(l+1)$ holdout is $\equiv 1 \pmod 3$, so its half is $\equiv 2 \pmod 3$ and its quarter is $\equiv 1 \pmod 3$; only the quarter can coincide with another level's own last-holdout class by Theorem 12 again. □

Theorem 15 (Mod-9 two-class bound)**.** *Writing* $J := j^*(l)$*,* $H(l, J-1) \bmod 9 \subseteq \{4^J, 4^{J+1}\} \pmod 9$*.*

Theorem 15 follows from Theorem 7's reduction modulo 9 (where 2 is a primitive root, so residue classes correspond to elements of $\mathbb{Z}/6$ via discrete logarithm) applied one step before extinction. Numerically, $H(l, J-1)$ occupies the single class 4^J at every level we computed, $l = 5,\dots,21$; the exclusion of the other logically possible class, 4^{J+1}, is equivalent to a single parity statement about the second-highest exponent of the extremal witness, and is the sharpest concrete open question this part of the investigation produced.

Proposition 16 (H-014 equivalence)**.** $\mathrm{maxrun}(H(l, l+1))$ *bounded by a constant* C*, for all* l*, is logically equivalent to* $j^*(l) \leq l + O(1)$*, not merely sufficient for it.*

Proof sketch. Sufficiency is Corollary 9 at $j = l+1$. For necessity, the same corollary read downward gives $\mathrm{maxrun}(H(l, l+1)) \geq j^*(l) - l - 1$ directly from the definition of maxrun as a witness to non-covering. □

We flag this equivalence because it corrects an initial, mistaken framing of our own: the naive expectation, that a random set of holdouts at the observed density would have doubling chains far longer than the flat ceiling of 3–4 we measured at every level $l = 5,\dots,22$, is wrong. Once the correct, rapidly declining density of $H(l, l+1)$ is used rather than its value at small l, the expected longest chain under a matched-density independence model tracks the observed sequence within about one unit at every level, including both the one rise and the one fall in the sequence. $\mathrm{maxrun}(H(l, l+1))$ is not anomalously small; it is statistically typical for the set's density. Proposition 16 converts this from a reassuring-looking numerical coincidence into the correct reading: bounding $\mathrm{maxrun}(H(l, l+1))$ is exactly as hard as the rate bound it would imply, not an easier route to it.

A genuine, well-defined reduction of the harder, converse direction remains. Writing $U(l, W)$ for the covering family at width W (Section 7's reparameterization), the exact identity $U(W+1) = U(W) \cup 2U(W) \cup \mathrm{Corner}(W+1)$ holds, where $\mathrm{Corner}(W+1)$ consists of the witnesses using both the top and bottom available exponent simultaneously. At every width $W \geq 2l+1$ we were able to check exhaustively, $l = 3,\dots,13$, $\mathrm{Corner}(W+1)$ contributes nothing new: every corner witness's value is already covered without it. If this *corner-redundancy* property holds at every sufficiently large width, in general, it would make $j^*(l)$ computable from a single holdout set via Corollary 9's converse, a real practical gain for extending Table 1 beyond $l = 23$; it remains unproven.

8. Discussion

Two named lemmas summarize what stands between this paper's results and a sharper theorem. Excluding the class 4^{J+1} in Theorem 15 would make the mod-9 single-class law, observed at every level checked, a proven statement rather than an observation. Proving corner-redundancy at every sufficiently large width, in the sense of Section 7, would make Corollary 9's bootstrap tight in general, not merely empirically exact at every computed instance, and would resolve Proposition 16's question in the process. Neither lemma is, on its face, as hard as the Weak Covering Conjecture itself; whether either is substantially easier is, honestly, not yet known.

Section 5's three findings are, we believe, of independent interest beyond this specific covering problem. A finite exponential sum built from strictly decreasing, exponentially weighted digit tuples is a natural object in several adjacent areas (finite-population L-statistics at non-CLT frequencies and weights, Littlewood–Offord theory for structured coefficient sequences, and the analytic side of the Collatz literature via Tao's own related discussion), and the specific mechanism identified here, that generic cancellation caps out below what is already provable by combinatorial means, that the exceptional mass does not concentrate on a sparse set at the scale that matters, and that the actual obstruction is an extremal, non-generic arithmetic phenomenon rather than a statistical one, may recur elsewhere in that literature under different names. We would welcome a reader who recognizes it.

The Weak Covering Conjecture itself, and the exact asymptotic rate of $e(l)$, remain open.

9. Code and data availability

The repository `https://github.com/faculdade/weak-covering-conjecture` is reserved for this paper's code and data, organized by section, and will be populated before submission.

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